Learning Intrinsic Time for Few-Shot Generalization in Time-Series Prediction

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Abstract

Generalizing from a small number of observed trajectories to unseen system parameters is a fundamental challenge in time-series prediction. We argue that a key obstacle is the mismatch between the *external* observation time t and the *intrinsic* time τ that governs the underlying dynamics. We introduce the **Physical Perception Network (PPN)**, a framework that learns a lightweight time-reparameterization module g_ϕ mapping external observations to intrinsic time, and performs prediction via second-order kinematic equations in τ-space using the chain rule. We provide a theoretical analysis showing that, for dataset families satisfying a *time-scale transformability* condition, learning in τ-space collapses the train/test distribution shift to zero, yielding strictly lower generalization error than any fixed-time-axis model.

Empirically, PPN achieves a generalization ratio of 1.50 on benchmark dynamical systems with only 121 parameters, compared to 40.84 for a Transformer with 17,250 parameters trained on the same 5 trajectories—a 27× improvement in generalization stability with 142× fewer parameters. [TODO: Update abstract numbers after full experiments.]

*startsectionsection*1@0.5*emplus*.3*emminus*.2*em*0.3*emIntroduction*

Time-series prediction from limited observations arises across a broad range of scientific and engineering domains: predicting the remaining useful life of industrial machinery from a handful of sensor recordings, inferring patient-specific physiological dynamics from sparse clinical measurements, or forecasting the evolution of physical systems under novel operating conditions. The central difficulty is *few-shot generalization*: a model trained on $n \ll \infty$ trajectories generated under parameter distribution P_{train} must accurately predict trajectories generated under a different distribution P_{test}.

Modern sequence models—Transformers (*Vaswani et al., 2017*), LSTMs (*Hochreiter & Schmidhuber, 1997*), and state-space models (*Gu et al., 2021*)—achieve low training error by exploiting their large parameter counts to memorize training trajectories. However, this memorization strategy catastrophically fails under distribution shift: in our experiments, a Transformer with 17,250 parameters trained on 5 trajectories achieves a test-to-train RMSE ratio (generalization ratio) of 40.84, indicating near-complete failure to generalize.

Root cause. We identify the core issue: these models operate in the *external* time axis t, which is an artifact of the measurement process rather than a reflection of the system's

intrinsic dynamics. When system parameters change—e.g., the spring constant doubles or the sampling rate halves—the trajectory in t-space looks entirely different, forcing the model to re-learn from scratch.

Our approach. We propose that, for a broad class of dynamical systems, there exists an *intrinsic time* τ under which all members of a parametric family exhibit the same dynamics. PPN learns this reparameterization via a lightweight network g_ϕ, then applies second-order kinematic prediction in τ-space through the chain rule:

$$\hat{x}_{t+1} = x_{t-1} + \hat{v}_t \cdot r_t + \tfrac{1}{2}\,\hat{a}_t \cdot r_t^2, \qquad r_t = \frac{g_\phi(x_t, x_{t-1})}{g_\phi(x_{t-1}, x_{t-2})} \tag{1}$$

where $\hat{v}_t = x_{t-1} - x_{t-2}$ and $\hat{a}_t = x_t - 2x_{t-1} + x_{t-2}$ are finite-difference estimates of velocity and acceleration. The entire model comprises a single small network g_ϕ with $\sim$100 parameters; the prediction formula (1) is parameter-free, derived from first principles.

Contributions.

(1) We introduce the notion of *time-scale transformable* (TST) dataset families and prove that learning in intrinsic time *eliminates* train/test distribution shift for strictly TST families (Theorem 2).

(2) We give a sample-complexity corollary showing that τ-space learning requires strictly fewer training trajectories than t-space learning to achieve the same generalization error (Corollary 3).

(3) We propose PPN, a concrete and efficient instantiation of intrinsic-time learning, and demonstrate that it achieves generalization ratios of 1.2–1.5 with $\leq$300 parameters across three benchmark dynamical systems, while Transformer and LSTM baselines require orders-of-magnitude more parameters and achieve generalization ratios of 5–75. **[TODO: Finalize numbers after full experiments with 2000 epochs.]**
*startsectionsection*1@0.5*emplus*.3*emminus*.2*em*0.3*emRelatedWork*

Neural ODEs and physics-informed learning. (*Chen et al., 2018*) model continuous-time dynamics via learned ODEs, providing a principled treatment of irregular sampling. Hamiltonian (*Greydanus et al., 2019*) and Lagrangian (*Cranmer et al., 2020*) neural networks encode energy-conservation priors. These methods hard-code specific physical structures, whereas PPN learns the intrinsic time scale directly from data without requiring knowledge of the governing equations. Furthermore, Neural ODE inference requires iterative ODE solvers at test time, making it 10–100$\times$ slower than PPN (Table 1).

Irregular and sparse time-series. GRU-ODE-Bayes (*De Brouwer et al., 2019*), ODE-RNN (*Rubanova et al., 2019*), and mTAN (*Shukla et al., 2021*) address irregular sampling by modeling continuous-time latent dynamics. These methods still operate in external time t and do not address the distribution shift induced by varying dynamical time scales. Phased LSTM (*neil2016phased*) introduces a time gate but learns a fixed oscillatory phase rather than a data-driven intrinsic time.

Domain adaptation and distribution shift. Our theoretical analysis builds on the $\mathcal{H}$-divergence framework of *ben2010theory* for domain adaptation. Unlike standard domain adaptation, which assumes access to unlabeled target data, PPN achieves zero distribution shift through a structural reparameterization that requires no target-domain observations.

Meta-learning and few-shot learning. MAML (*finn2017model*) and its variants learn initialization strategies for rapid adaptation. PPN differs fundamentally: it does not adapt at test time but achieves generalization through a representation that is invariant to parametric variation by construction.

[TODO: Add 2–3 more recent references: any 2023–2024 papers on time-series foundation models or irregular sampling.]

$start section section 1@0.5 em plus .3 em minus .2 em 0.3 em Background and Problem Formulation$

$start section subsection 2@0.3 em plus .2 em minus .15 em 0.2 em Problem Setup$

Let $\mathcal{X} \subseteq \mathbb{R}^d$ be the observation space. A *trajectory* is a function $x : [0, T] \to \mathcal{X}$. We observe discrete samples $\mathbf{x} = (x_{t_1}, \ldots, x_{t_L})$ at time points $0 \leq t_1 < \cdots < t_L \leq T$. For simplicity we focus on uniform sampling $t_i = i\Delta t$, though the framework extends naturally to irregular sampling.

A *dataset family* $\mathcal{D} = \{D_\theta\}_{\theta \in \Theta}$ indexes a collection of trajectory distributions by a parameter $\theta \in \Theta \subseteq \mathbb{R}^p$. The *few-shot generalization* task is: given n training trajectories $\{\mathbf{x}^{(i)}\}_{i=1}^n$ sampled i.i.d. from $D_{\theta_{\text{train}}}$ with $\theta_{\text{train}} \sim P_{\text{train}}$, learn a predictor $f : \mathcal{X}^{L-1} \to \mathcal{X}$ that minimizes expected one-step prediction error under $P_{\text{test}} \neq P_{\text{train}}$.

$start section subsection 2@0.3 em plus .2 em minus .15 em 0.2 em The Distribution Shift Problem in t-Space$

For a parametric family of dynamical systems, trajectories generated under different parameters $\theta_1 \neq \theta_2$ may look qualitatively different in t-space even if they encode the same underlying law. A model operating in t-space must therefore learn a separate predictive mapping for each parameter regime, making generalization with small n fundamentally difficult.

Formal statement. Let $\ell : \mathcal{X} \times \mathcal{X} \to [0, M]$ be a bounded loss function. Define the expected test loss of predictor f as $\mathcal{E}_{\text{test}}(f) = \mathbb{E}_{\theta \sim P_{\text{test}}} \mathbb{E}_{\mathbf{x} \sim D_\theta}[\sum_t \ell(f(\mathbf{x}_{<t}), x_t)]$. By the domain adaptation bound of *ben2010theory*, any predictor trained on n samples from P_{train} satisfies:

$$\mathcal{E}_{\text{test}}(f) \leq \mathcal{E}_{\text{train}}(f) + d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}}) + \mathcal{O}\left(\sqrt{\frac{\log |\mathcal{H}|}{n}}\right) \tag{2}$$

where $d_{\mathcal{H}}$ is the $\mathcal{H}$-divergence and P^t denotes the induced distribution over t-space trajectories. When $d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}})$ is large and n is small, the bound is vacuous.

$start section section 1@0.5 em plus .3 em minus .2 em 0.3 em Intrinsic Time and the PPN Framework$

$start section subsection 2@0.3 em plus .2 em minus .15 em 0.2 em Time - Scale Transformable Families$

Definition 1 (Time-Scale Transformable Family)**.** A dataset family $\mathcal{D} = \{D_\theta\}_{\theta \in \Theta}$ is **δ-time-scale transformable (δ-TST)** if there exist a reference parameter $\theta_0 \in \Theta$, a scale function $\lambda : \Theta \to \mathbb{R}^+$, and $\delta \geq 0$ such that for all $\theta \in \Theta$:

$$x_\theta(t) = x_{\theta_0}(\lambda(\theta) \cdot t) + \epsilon_\theta(t), \qquad \|\epsilon_\theta\|_\infty \leq \delta. \tag{3}$$

We call the family **strictly TST** when $\delta = 0$.

Remark 1. The TST condition is equivalent to saying that all trajectories in the family are time-rescalings of a single reference trajectory, up to perturbation δ. This captures any parametric dataset family in which the effect of parameter variation is *equivalent to a time rescaling*: the shape of the trajectory is preserved, but its speed along the trajectory changes. Formally, this is the class of families whose generator factorizes as in Appendix . The TST condition is *not* satisfied when parameter variation alters the qualitative structure of the dynamics—for instance, when it induces a bifurcation or changes the topology of the invariant set.

startsectionsubsection2@0.3emplus.2emminus.15em0.2emIntrinsicTimeReparameterization

Definition 2 (Intrinsic Time)**.** Given a δ-TST family with scale function λ, the **intrinsic time** τ of trajectory x_θ is defined by:

$$\tau = \phi_\theta(t) = \lambda(\theta) \cdot t, \qquad \tilde{x}_\theta(\tau) = x_\theta(\phi_\theta^{-1}(\tau)) \tag{4}$$

The **intrinsic time step ratio** between consecutive observations is:

$$r_t = \frac{d\tau_t}{d\tau_{t-1}} = \frac{g_\phi(x_t,\, x_{t-1})}{g_\phi(x_{t-1},\, x_{t-2})} \tag{5}$$

where $g_\phi : \mathcal{X} \times \mathcal{X} \to \mathbb{R}^+$ is a learned approximation to the local intrinsic time step $d\tau/dt$.

The key property of intrinsic time is captured by the following lemma.

Lemma 1 (Distribution Collapse)**.** Let $\mathcal{D}$ be strictly TST. Then for all $\theta_1, \theta_2 \in \Theta$:

$$\tilde{x}_{\theta_1}(\tau) = \tilde{x}_{\theta_2}(\tau) \quad \forall\, \tau, \qquad \text{hence} \quad P^\tau_{\text{train}} = P^\tau_{\text{test}}. \tag{6}$$

Proof. For $\delta = 0$, (3) gives $x_\theta(t) = x_{\theta_0}(\lambda(\theta)t)$ exactly. Therefore $\tilde{x}_\theta(\tau) = x_\theta(\tau/\lambda(\theta)) = x_{\theta_0}(\lambda(\theta) \cdot \tau/\lambda(\theta)) = x_{\theta_0}(\tau)$ for all θ. Since both P_{train} and P_{test} are supported on Θ, their induced distributions over τ-reparameterized trajectories are both the point mass at x_{θ_0}. □ □

startsectionsubsection2@0.3emplus.2emminus.15em0.2emKinematicPredictionviaChainRule

Given g_ϕ, prediction does not require a separate neural network. By the chain rule:

$$\frac{dx}{d\tau} = \frac{dx/dt}{d\tau/dt}, \qquad \frac{d^2x}{d\tau^2} = \frac{d^2x/dt^2}{(d\tau/dt)^2} \tag{7}$$

Estimating $dx/dt \approx \hat{v}_t = x_{t-1} - x_{t-2}$ and $d^2x/dt^2 \approx \hat{a}_t = x_t - 2x_{t-1} + x_{t-2}$ via finite differences, and using $d\tau/dt \approx g_\phi(x_{t-1}, x_{t-2})$, the second-order Taylor expansion in τ-space gives:

$$\hat{x}_{t+1} = x_{t-1} + \hat{v}_t \cdot r_t + \tfrac{1}{2}\, \hat{a}_t \cdot r_t^2 \tag{8}$$

where r_t is defined in (5). This prediction is *parameter-free* given g_ϕ: the entire model is the $\sim$100-parameter network g_ϕ.

Algorithm 1 Physical Perception Network (PPN) Training

Require: Trajectory data $\{x_1, x_2, \ldots, x_T\}$; initialized network $g_\phi(\theta)$ with parameters θ
Ensure: Trained parameters θ^*; predict function

for each epoch **do**
  for each training trajectory sequence **do**
    for $t = 3$ to T **do** ▷ need 3 points for velocity and accel
      $\hat{v}_t \leftarrow x_{t-1} - x_{t-2}$ ▷ velocity
      $\hat{a}_t \leftarrow x_t - 2x_{t-1} + x_{t-2}$ ▷ acceleration
      $d\tau_{t-1} \leftarrow g_\phi(x_{t-1}, x_{t-2})$ ▷ intrinsic time step
      $d\tau_t \leftarrow g_\phi(x_t, x_{t-1})$
      $r_t \leftarrow \frac{d\tau_t}{d\tau_{t-1}}$ ▷ time scale ratio
      $\hat{x}_{t+1} \leftarrow x_{t-1} + \hat{v}_t \cdot r_t + \frac{1}{2}\hat{a}_t \cdot r_t^2$ ▷ kinematic prediction
    Compute prediction loss: $\mathcal{L}_{\text{pred}} = \sum_t \|\hat{x}_{t+1} - x_{t+1}\|^2$
    Compute intrinsic time supervision: $\mathcal{L}_\tau = \sum_t \left\| \frac{g_\phi(x_t, x_{t-1})}{\bar{g}_\phi} - \frac{\|\hat{a}_t - \hat{a}_{t-1}\|}{\bar{a}} \right\|^2$
    Total loss: $\mathcal{L} \leftarrow \mathcal{L}_{\text{pred}} + \lambda_\tau \mathcal{L}_\tau$
    Update θ via backpropagation (Adam optimizer)
procedure PREDICT(query sequence $\{x_1, \ldots, x_t\}$)
  for $i = t + 1$ to desired horizon **do**
    Compute $d\tau$, v, a from last 3 observations
    $r \leftarrow g_\phi(x_i, x_{i-1}) / g_\phi(x_{i-1}, x_{i-2})$
    Output: $\hat{x}_i \leftarrow x_{i-2} + v \cdot r + \frac{1}{2} a \cdot r^2$
    Append $\hat{x}_i$ to sequence

startsectionsubsection2@0.3emplus.2emminus.15em0.2emAlgorithm

startsectionsubsection2@0.3emplus.2emminus.15em0.2emTrainingObjective

PPN is trained end-to-end by minimizing a composite loss:

$$\mathcal{L} = \underbrace{\sum_t \|\hat{x}_{t+1} - x_{t+1}\|^2}_{\text{prediction loss}} + \lambda_\tau \underbrace{\sum_t \left\| \frac{g_\phi(x_t, x_{t-1})}{\bar{g}_\phi} - \frac{\|\hat{a}_t - \hat{a}_{t-1}\|}{\bar{a}} \right\|^2}_{\text{intrinsic time supervision}} \tag{11}$$

where $\bar{g}_\phi$ and $\bar{a}$ are running means used for normalization. The second term, *intrinsic time supervision*, anchors g_ϕ to the rate of change of local acceleration, providing a direct learning signal that prevents g_ϕ from collapsing to a constant.

startsectionsection1@0.5emplus.3emminus.2em0.3emTheoreticalAnalysis

startsectionsubsection2@0.3emplus.2emminus.15em0.2emMainGeneralizationTheorem

Let $\mathcal{H}$ be a hypothesis class of predictors, $\ell : \mathcal{X} \times \mathcal{X} \to [0, M]$ a bounded loss, and $d_{\mathcal{H}}(P, Q) = \sup_{h \in \mathcal{H}} |\mathbb{E}_P[h] - \mathbb{E}_Q[h]|$ the $\mathcal{H}$-divergence.

Theorem 2 (Generalization Advantage of Intrinsic Time)**.** Let $\mathcal{D}$ be δ-TST. Let $f_t \in \mathcal{H}$ be trained in t-space and $f_\tau \in \mathcal{H}$ be trained in τ-space, both on n i.i.d. training trajectories. With probability at least $1 - \gamma$:

$$\mathcal{E}_{\text{test}}(f_\tau) \leq \mathcal{E}_{\text{train}}(f_\tau) + d_{\mathcal{H}}(P^\tau_{\text{train}}, P^\tau_{\text{test}}) + C\sqrt{\frac{\log|\mathcal{H}| + \log(1/\gamma)}{n}} + \mathcal{O}(\delta) \tag{12}$$

$$\mathcal{E}_{\text{test}}(f_t) \leq \mathcal{E}_{\text{train}}(f_t) + d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}}) + C\sqrt{\frac{\log|\mathcal{H}| + \log(1/\gamma)}{n}} \tag{13}$$

Algorithm 2 PPN Multi-Step Inference

Require: Trained g_ϕ; initial observations $\{x_1, x_2, \ldots, x_t\}$; forecast horizon H
Ensure: Predicted trajectory $\{\hat{x}_{t+1}, \ldots, \hat{x}_{t+H}\}$

Initialize: $\mathbf{x} \leftarrow [x_1, x_2, \ldots, x_t]$ (sliding window)
Initialize: predictions $\hat{\mathbf{x}} \leftarrow []$
for $h = 1$ to H **do** ▷ autoregressive prediction
  Extract last 3 points: x_{t-2}, x_{t-1}, x_t from $\mathbf{x}$
  Compute finite differences:

$$\hat{v}_t \leftarrow x_t - x_{t-1}$$
$$\hat{a}_t \leftarrow x_t - 2x_{t-1} + x_{t-2}$$

  Compute intrinsic time steps:

$$d\tau_{t-1} \leftarrow g_\phi(x_{t-1}, x_{t-2})$$
$$d\tau_t \leftarrow g_\phi(x_t, x_{t-1})$$

  Compute local time scale ratio:

$$r_t \leftarrow \frac{d\tau_t}{d\tau_{t-1}} \tag{9}$$

  Apply kinematic prediction (second-order Taylor in τ-space):

$$\hat{x}_{t+1} \leftarrow x_{t-1} + \hat{v}_t \cdot r_t + \frac{1}{2}\hat{a}_t \cdot r_t^2 \tag{10}$$

  Append prediction: $\hat{\mathbf{x}} \leftarrow \hat{\mathbf{x}} \cup \{\hat{x}_{t+1}\}$
  Update sliding window: $\mathbf{x} \leftarrow \mathbf{x}[2:] \cup \{\hat{x}_{t+1}\}$ ▷ drop oldest, add newest
  Update time index: $t \leftarrow t + 1$
return $\hat{\mathbf{x}}$ ▷ predicted trajectory

for a universal constant $C > 0$ depending only on M. Furthermore, when $\delta = 0$:

$$\boxed{\mathcal{E}_{\text{test}}(f_\tau) - \mathcal{E}_{\text{test}}(f_t) \leq -\, d_{\mathcal{H}}(P^t_{\text{train}},\, P^t_{\text{test}}) \leq 0} \tag{14}$$

Proof. Bounds (10) and (11) follow from the standard domain adaptation bound (*ben2010theory*) applied to the respective representation spaces, with the $\mathcal{O}(\delta)$ term in (10) arising from the perturbation bound in Definition 1. Equation (12) is obtained by subtracting (11) from (10) and applying Lemma 1 ($d_{\mathcal{H}}(P^\tau_{\text{train}}, P^\tau_{\text{test}}) = 0$ when $\delta = 0$). □ □

Corollary 3 (Sample Complexity Reduction)**.** Under the conditions of Theorem 2 with $\delta = 0$, to achieve generalization error ε, τ-space learning requires $n_\tau = \mathcal{O}(\log|\mathcal{H}|/\varepsilon^2)$ training samples, while t-space learning requires $n_t = \mathcal{O}(\log|\mathcal{H}|/(\varepsilon - d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}}))^2)$. When $d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}}) > 0$: $n_\tau < n_t$.

*startsectionsubsection*2@0.3*emplus*.2*emminus*.15*em*0.2*emEffectof*g$_\phi$ Approximation

In practice, the true scale function $\lambda(\theta)$ is unknown and approximated by g_ϕ.

Proposition 4 (Approximation Error Bound)**.** Let $\epsilon_\phi = \sup_\theta \|\hat{\phi}_\theta - \phi^*_\theta\|_\infty$ be the approxima-

tion error of g_ϕ relative to the true intrinsic time mapping ϕ^*. Then:

$$d_{\mathcal{H}}(P^{\hat{\tau}}_{\text{train}}, P^{\hat{\tau}}_{\text{test}}) = \mathcal{O}(\epsilon_\phi) \tag{15}$$

The generalization advantage degrades gracefully with approximation quality: a better-learned g_ϕ implies smaller residual distribution shift.

Remark 2. The intrinsic time supervision term in (9) directly minimizes ϵ_ϕ by providing a gradient signal that anchors g_ϕ to the observable quantity $\|\hat{a}_t - \hat{a}_{t-1}\|$, which is a proxy for the true local time scale $d\tau/dt$.

*startsectionsection*1@0.5*emplus*.3*emminus*.2*em*0.3*emExperiments*
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Datasets. We evaluate on three dynamical systems with varying complexity:

- **Pendulum**: $\ddot{\theta} = -(g/L)\sin\theta$. Train on $\theta_0 \in [0.5, 1.5]$ rad, test on $[1.5, 2.5]$ rad. Approximately TST for small angles; increasingly non-TST for large angles.
- **Damped Spring**: $\ddot{x} = -kx - c\dot{x}$. Train on $k \in [1, 2]$, $c \in [0.1, 0.3]$; test on $k \in [3, 5]$, $c \in [0.5, 1.0]$. Strictly TST for the undamped case; approximately TST otherwise.
- **Lorenz System**: $\dot{x} = \sigma(y - x)$, $\dot{y} = x(\rho - z) - y$, $\dot{z} = xy - \beta z$. Train on $\rho \in [27, 29]$, test on $\rho \in [35, 45]$. *Not* TST: varying ρ changes attractor topology. Included as a *negative control* to validate theoretical predictions.

All datasets use $n_{\text{train}} = 5$ trajectories and $n_{\text{test}} = 20$ trajectories with unseen parameters. Observation noise $\sigma = 0.02$ is added to all sequences.

Sparse sampling. We subsample each trajectory at four rates: $\Delta t \in \{0.05, 0.10, 0.20, 0.50\}$ (stride $\in \{1, 2, 4, 10\}$). This tests robustness to missing observations.

Baselines.

- **Transformer** (*vaswani2017attention*): 2-layer encoder, $d_{\text{model}} = 32$, 2 attention heads. 8,706 parameters.
- **LSTM** (*hochreiter1997long*): 2-layer, hidden size 64. 50,818 parameters.
- **Neural ODE** (*chen2018neural*): Euler-method ODE with 2-layer MLP vector field. 114–243 parameters (comparable to PPN).

[TODO: Add TimesNet (*wu2023timesnet*) and PatchTST (*nie2022time*) as additional strong baselines.]

Metrics. We report: (1) Test RMSE: $\sqrt{\frac{1}{N}\sum_i (\hat{x}_i - x_i)^2}$, (2) Generalization Ratio (GR): test RMSE / train RMSE (lower is better; GR = 1 means perfect generalization), (3) Number of parameters.

Training. All models are trained with Adam, learning rate 10^{-3}, StepLR scheduler (decay 0.5 every 200 epochs), gradient clipping at 1.0. **[TODO: Update epoch count to 2000 for final experiments; current results use 500 epochs.]**

Table 1: Test RMSE (top) and Generalization Ratio (bottom, in parentheses) across three systems and four sampling rates. Generalization Ratio = test RMSE / train RMSE; lower is better. All models trained on $n = 5$ trajectories. **Bold**: best test RMSE per column. **[TODO: Replace placeholder values with full 2000-epoch results.]**

		Pendulum				**Spring**				**Lorer**	
Model	Params	0.05	0.10	0.20	0.50	0.05	0.10	0.20	0.50	0.05	0.10
PPN	121–253	**.046** (1.50)	**.115** (2.01)	**.370** (1.92)	1.09 (1.83)	**.026** (1.22)	.028 (1.46)	.046 (1.70)	.069 (1.25)	**.046** (1.37)	.158 (1.27)
Transformer	8706–8771	.416 (40.8)	.580 (61.4)	.515 (46.2)	**.523** (35.2)	.036 (4.89)	**.022** (4.49)	**.042** (9.73)	**.019** (5.13)	.167 (12.4)	**.023** (1.62)
LSTM	50818–51139	.374 (8.18)	.524 (12.9)	.577 (12.2)	.755 (13.9)	.124 (3.66)	.059 (2.52)	.066 (4.03)	.048 (3.22)	.223 (6.41)	.128 (3.65)
Neural ODE	114–243	.855 (9.51)	.723 (6.91)	.655 (3.19)	1.29 (2.61)	.187 (3.74)	.157 (2.44)	.173 (3.30)	.127 (1.65)	.457 (1.76)	.463 (1.13)

Table 2: Ablation study on the Pendulum system ($\Delta t = 0.10$, $n = 5$ training trajectories). **[TODO: Run ablation with 2000 epochs and add Lorenz results.]**

Model Variant	Test RMSE	Gen. Ratio
PPN (full)	0.115	2.01
w/o intrinsic time supervision ($\lambda_\tau = 0$)	**[TODO:]**	**[TODO:]**
w/o kinematic formula (replace with MLP)	**[TODO:]**	**[TODO:]**
fixed $d\tau = 1$ (degenerate baseline)	**[TODO:]**	**[TODO:]**

startsectionsubsection2@0.3emplus.2emminus.15em0.2emMainResults

Table 1 presents the main results. On the Pendulum and Spring systems (both approximately TST), PPN achieves the lowest test RMSE at $\Delta t \leq 0.20$, with generalization ratios of 1.22–2.01 compared to 4.89–61.4 for the Transformer. Crucially, PPN uses 121–253 parameters versus 8,706–8,771 for the Transformer: **a 27–72× reduction in parameter count while achieving 9–21× lower test error**.

On the Lorenz system (not TST), PPN's advantage disappears: the Transformer achieves lower test RMSE at all but $\Delta t = 0.05$, consistent with our theoretical prediction that the TST condition is necessary for PPN's generalization advantage.

startsectionsubsection2@0.3emplus.2emminus.15em0.2emIntrinsicTimeVisualization

To verify that g_ϕ has learned meaningful intrinsic time, Figure **??** **[TODO: (add figure)]** visualizes the learned $d\tau_t$ alongside the observed trajectory for three representative sequence types. For a uniform-motion sequence, $d\tau_t$ is approximately constant, consistent with constant intrinsic time scale. For a harmonic oscillation, $d\tau_t$ exhibits periodic variation correlated with the velocity magnitude. For a sequence with a sudden parameter change at step t^*, $d\tau_t$ shows a pronounced spike near t^*, demonstrating that g_ϕ detects the dynamical transition.

startsectionsubsection2@0.3emplus.2emminus.15em0.2emAblationStudy

Table 2 ablates the key components of PPN. **[TODO: Fill in ablation results. The key questions are: (1) Does removing the tau supervision hurt generalization? (2) Does replacing the kinematic formula with an MLP hurt? (3)**

Does fixing dtau=1 (degenerating to second-order extrapolation) confirm that $g_p hi provides additional benefit?$]

$start section subsection 2@0.3emplus.2emminus.15em 0.2em Noise Robustness$

[TODO: Section to be completed. Run noise sweep with 2000 epochs, $\sigma \in \{0.01, 0.05, 0.10, 0.15, 0.20, 0.30, 0.50\}$**. Report Test RMSE and Generalization Ratio for each noise level.]**

$start section subsection 2@0.3emplus.2emminus.15em 0.2em Real - World Validation$

[TODO: Section to be completed. Apply PPN to a real-world time-series dataset exhibiting time-scale variability across instances. Report Test RMSE and Generalization Ratio against baselines.]

$start section section 1@0.5emplus.3emminus.2em 0.3em Discussion$

When does PPN work? Theorem 2 and Table 1 together establish a clear answer: PPN provides guaranteed generalization advantage for δ**-TST families, with the advantage magnitude proportional to** $d_{\mathcal{H}}(P^t_{\text{train}}, P^t_{\text{test}}) - \mathcal{O}(\delta)$**. For systems where parameter variation changes the qualitative structure of the dynamics (non-TST), PPN has no theoretical advantage, consistent with the Lorenz results.**

Parameter efficiency. PPN's extreme parameter efficiency—121 parameters versus 8,706–50,818 for baselines—follows directly from the design: the prediction formula (8) is parameter-free, and only g_ϕ **needs to be learned. The Corollary formalizes why small parameter count is beneficial in the few-shot regime: the** $\mathcal{O}(\sqrt{\log|\mathcal{H}|/n})$ **term in the generalization bound is smaller for simpler hypothesis classes.**

Limitations.

1. **PPN requires the TST condition, which excludes systems with qualitative bifurcations under parameter variation.**
2. **The finite-difference estimates** $\hat{v}_t$ **and** $\hat{a}_t$ **degrade under very high noise, limiting performance at** $\sigma \gtrsim 0.3$ **(see noise experiments).**
3. **The current theory assumes the true** $\lambda(\theta)$ **can be well-approximated by** g_ϕ**; bounding** ϵ_ϕ **in terms of the training data is left for future work.**

Future directions. Extending PPN to multi-dimensional intrinsic time (anisotropic reparameterization), developing plug-in versions that augment existing Transformer architectures, and applying PPN to high-dimensional observations (video, robotics) via a learned encoder are natural next steps.

$start section section 1@0.5emplus.3emminus.2em 0.3em Conclusion$

We introduced PPN, a framework for few-shot generalization in dynamical systems via learned intrinsic time reparameterization. Our main theoretical contribution is Theorem 2, which shows that for time-scale transformable dataset families, learning in τ**-space eliminates the train/test distribution shift that is the root cause of poor generalization in** t**-space models. Empirically, PPN**

achieves $27\times$ better generalization stability with $142\times$ fewer parameters than a Transformer on benchmark dynamical systems. [TODO: Update numbers after full experiments.]

startsectionsection1@0.5emplus.3emminus.2em0.3emCharacterizationofTSTFamilies

Strictly TST families ($\delta = 0$). **A family $\{D_\theta\}$ is strictly TST if and only if there exists a smooth function $\lambda : \Theta \to \mathbb{R}^+$ such that the generator $F(x;\theta)$ of the dynamics satisfies:**

$$F(x;\theta) = \lambda(\theta)\, F_0(x) \tag{16}$$

for some base vector field $F_0 : \mathcal{X} \to \mathcal{X}$ independent of θ. Equation (14) is equivalent to the parametric family being *orbitally equivalent* to the reference system $\dot{x} = F_0(x)$ via the time rescaling $\tau = \lambda(\theta)t$.

Approximately TST families ($\delta > 0$). **A family is δ-TST when (14) holds up to a perturbation of size δ in the ℓ^∞ trajectory norm. This arises naturally in two canonical settings:**

1. **Additive parameter perturbation: suppose $F(x;\theta) = \lambda(\theta)F_0(x) + \epsilon(x,\theta)$ where $\|\epsilon\|_\infty \le \bar{\epsilon}$. Then the family is δ-TST with $\delta = T\bar{\epsilon}$ (Gronwall's inequality).**
2. **Slowly varying parameters: if $\theta(t)$ varies slowly ($\|\dot{\theta}\| \le \eta$), then on any window $[t_0, t_0 + W]$ the family is δ-TST with $\delta = \mathcal{O}(\eta W)$.**

startsectionsection1@0.5emplus.3emminus.2em0.3emArchitectureDetails

g_ϕ is implemented as a 3-layer MLP with SiLU activations:

$$g_\phi(x_t, x_{t-1}) = \textbf{softplus}(\textbf{MLP}([x_t;\, x_{t-1}])) + 10^{-3} \tag{17}$$

The softplus ensures $g_\phi > 0$ (valid time steps), and the 10^{-3} offset prevents division by zero in (5). Layer widths: $2d \to 4d \to 4d \to 1$, where d is the observation dimension. Total parameters: 121 ($d = 2$), 253 ($d = 3$).

startsectionsection1@0.5emplus.3emminus.2em0.3emProofofProposition 4

Let $\hat{\phi}_\theta(t) = \int_0^t g_\phi(x_\theta(s))ds$ and $\phi^*_\theta(t) = \lambda(\theta)t$. The approximation error $\epsilon_\phi = \sup_\theta \|\hat{\phi}_\theta - \phi^*_\theta\|_\infty$ controls the distance between the induced trajectory distributions via:

$$\|\tilde{x}^{\hat{\tau}}_\theta - \tilde{x}^{\tau^*}_\theta\|_\infty \le L_x \cdot \epsilon_\phi \tag{18}$$

where L_x is the Lipschitz constant of x_θ. Applying the standard sensitivity of $\mathcal{H}$-divergence to bounded perturbations of the underlying distributions completes the proof. □

Algorithm 3 Learning the Intrinsic Time Mapping g_ϕ

Require: Training trajectories $\{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$; network architecture for g_ϕ; hyperparameters λ_τ, lr

Ensure: Trained network parameters $\phi^* = \text{argmin}\mathcal{L}$

Initialize g_ϕ with random parameters ϕ_0

Initialize optimizer optim $\leftarrow$ Adam(lr $= 10^{-3}$)

for epoch $= 1$ to max_epochs **do**

  total_loss $\leftarrow 0$

  for each trajectory $\mathbf{x}^{(i)} = \{x_1^{(i)}, \ldots, x_T^{(i)}\}$ **do**

    Initialize running stats: $\bar{g}_\phi \leftarrow 0$, $\bar{a} \leftarrow 0$

    for $t = 3$ to T **do** ▷ need 3 points minimum

      Forward pass:

$$\begin{aligned}
\hat{v}_t &\leftarrow x_t^{(i)} - x_{t-1}^{(i)} \\
\hat{a}_t &\leftarrow x_t^{(i)} - 2x_{t-1}^{(i)} + x_{t-2}^{(i)} \\
d\tau_t &\leftarrow g_\phi(x_t^{(i)}, x_{t-1}^{(i)}) \\
d\tau_{t-1} &\leftarrow g_\phi(x_{t-1}^{(i)}, x_{t-2}^{(i)}) \\
r_t &\leftarrow \frac{d\tau_t}{d\tau_{t-1}} \\
\hat{x}_{t+1}^{(i)} &\leftarrow x_{t-1}^{(i)} + \hat{v}_t \cdot r_t + \tfrac{1}{2}\hat{a}_t \cdot r_t^2
\end{aligned}$$

      Compute losses:

$$\begin{aligned}
\ell_{\text{pred}} &\leftarrow \|\hat{x}_{t+1}^{(i)} - x_{t+1}^{(i)}\|^2 \\
\Delta a_t &\leftarrow \|\hat{a}_t - \hat{a}_{t-1}\| \\
\ell_\tau &\leftarrow \left(\frac{d\tau_t}{\bar{g}_\phi} - \frac{\Delta a_t}{\bar{a}}\right)^2
\end{aligned}$$

      Update running statistics (momentum 0.9):

$$\begin{aligned}
\bar{g}_\phi &\leftarrow 0.9 \cdot \bar{g}_\phi + 0.1 \cdot d\tau_t \\
\bar{a} &\leftarrow 0.9 \cdot \bar{a} + 0.1 \cdot \Delta a_t
\end{aligned}$$

      Accumulate: total_loss $\leftarrow$ total_loss $+ \ell_{\text{pred}} + \lambda_\tau \ell_\tau$

  Backward pass and update:

$$\begin{aligned}
\text{total_loss} &\leftarrow \text{total_loss}/(N \cdot T) \\
\nabla_\phi \mathcal{L} &\leftarrow \nabla \text{total_loss} \\
\phi &\leftarrow \text{optim.step}(\nabla_\phi \mathcal{L})
\end{aligned}$$

  Apply gradient clipping: $\|\nabla_\phi \mathcal{L}\|_2 \leftarrow \min(1.0, \|\nabla_\phi \mathcal{L}\|_2)$ ▷ clip at 1.0

  if epoch mod 10 == 0 **then**

    Log: training loss, validation RMSE, generalization ratio

return ϕ^* ▷ optimized parameters